Properties of the Region of Convergence

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Key Concepts

1) This lesson explores seven basic properties of the region of convergence (ROC).

2) The ROC depends only on $|z|$ and thus has three possibilities.

    a) The ROC is the exterior of a circle in the z-plane.

    b) The ROC is the interior of a circle in the z-plane.

    c) The ROC is a ring in the z-plane.

3) The ROC cannot contain any poles.

4) If the discrete-time Fourier transform exists, then the ROC includes the unit circle ($|z| = 1$).

5) If the signal is finite duration, then the ROC is all values of z except possibly $z = 0$ and/or $z = \infty$.

6) A right-sided signal is zero for $n < N_R$ and of infinite duration for positive n. The ROC of a right-sided signal is the exterior of a circle whose radius is given by the largest magnitude of the poles.

7) A left-sided signal is zero for $n > N_L$ and of infinite duration for negative n. The ROC of a left-sided signal is the interior of a circle whose radius is given by the smallest magnitude of the poles.

8) A two-sided signal is of infinite duration in both positive and negative n directions. The ROC of a two-sided signal lies in a ring bounded on the inside and outside by poles.

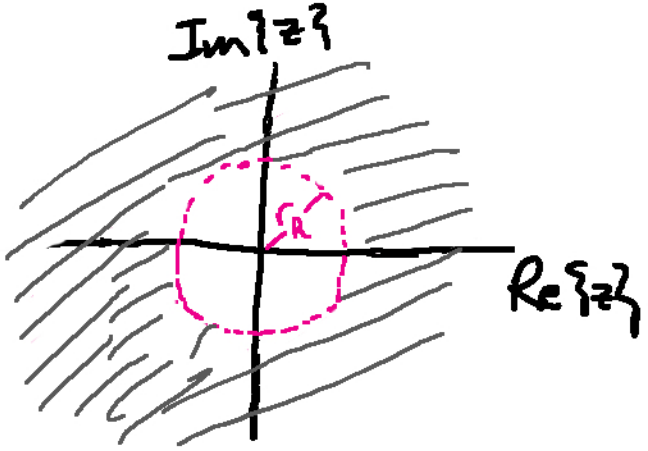
9) Some two-sided signals do not have a ROC. That is, their z-transform does not exist.

Video Screenshots

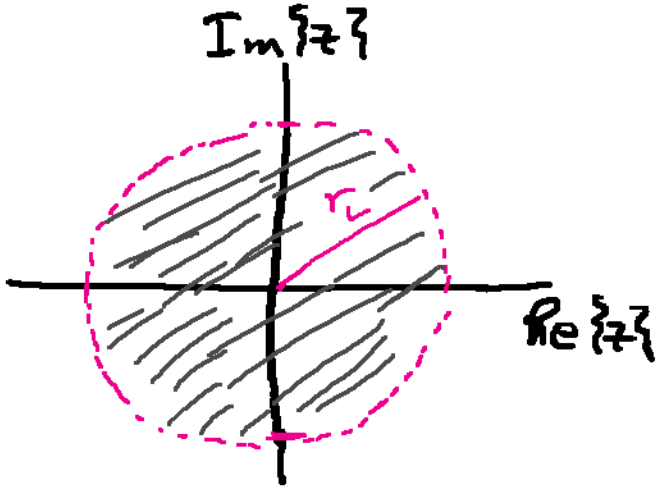
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Properties of the ROC

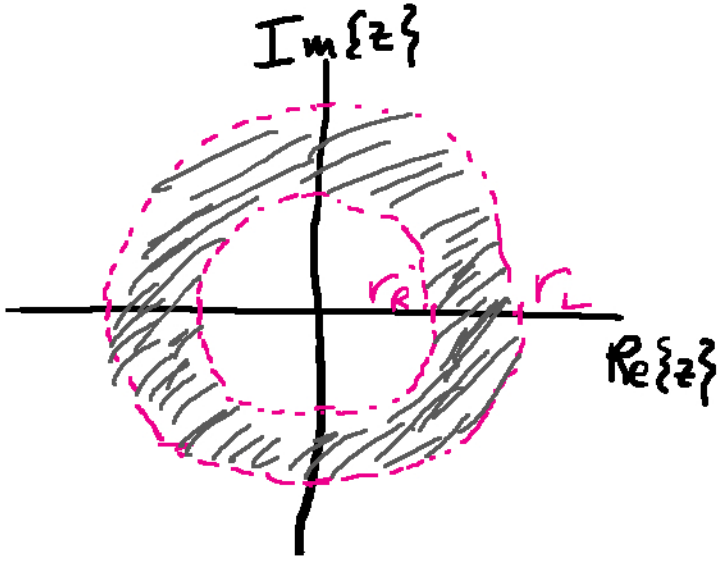
1) Three possibilities



$|z| > r_R$



$|z| < r_L$



$r_R < |z| < r_L$

2) ROC cannot contain any poles.

3) DTFT exists (absolutely summable)

$\Longleftrightarrow$ ROC includes $|z| = 1$ → unit circle

4) If $x[n]$ is finite duration $\Rightarrow$

$$x[n] = 0 \quad , \quad n < N_1 \; , \; n > N_2$$

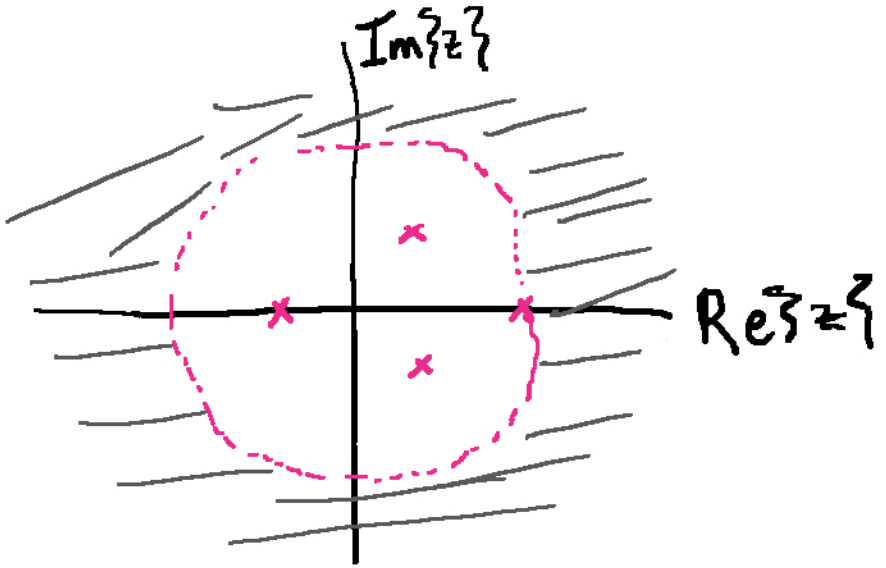
then ROC is all z except possibly $z = 0, \infty$

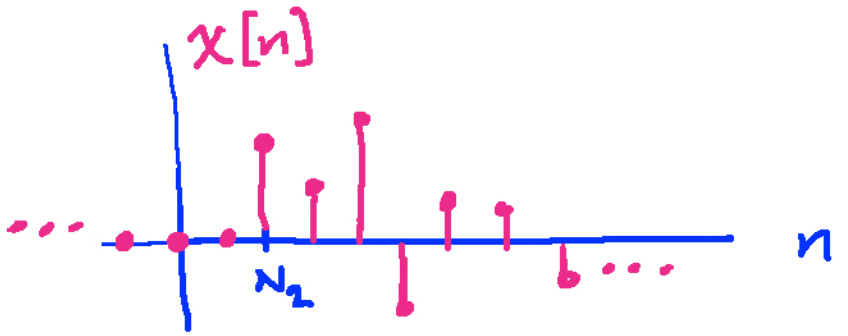
$$\sum_{n=N_1}^{N_2} |x[n] z^{-n}| < \infty$$

$z \neq 0$ if $N_2 > 0$

$z \neq \infty$ if $N_1 < 0$

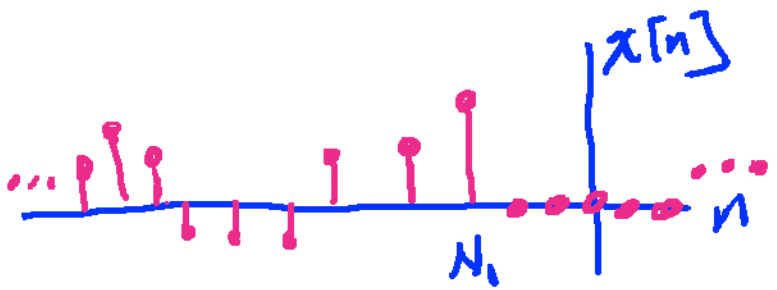
5) Right-sided signal

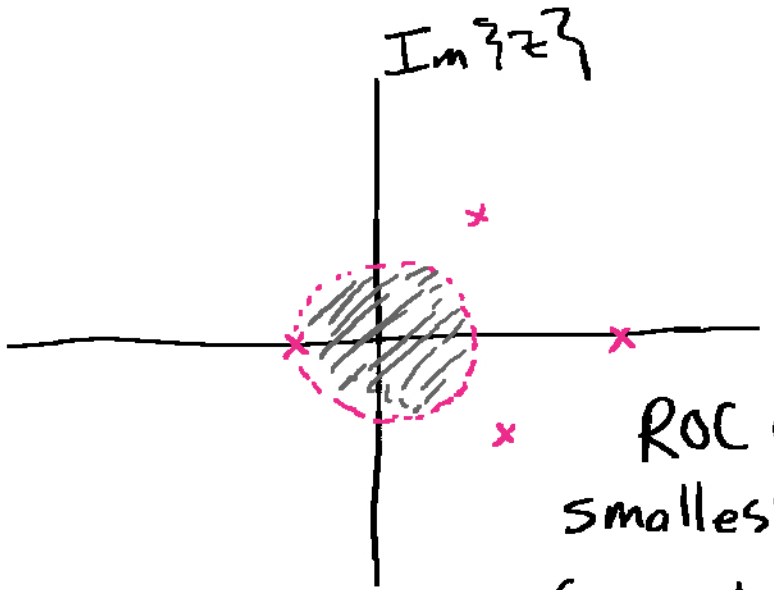




ROC extends out from largest magnitude pole (except possibly $z = \infty$)

6) Left-sided signal

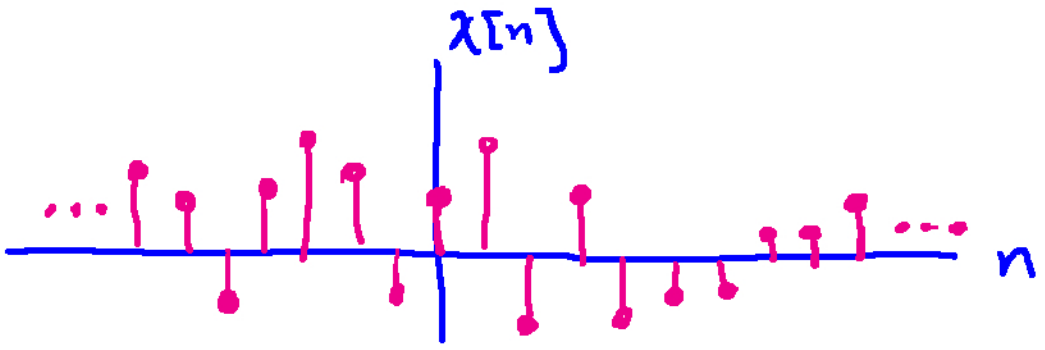


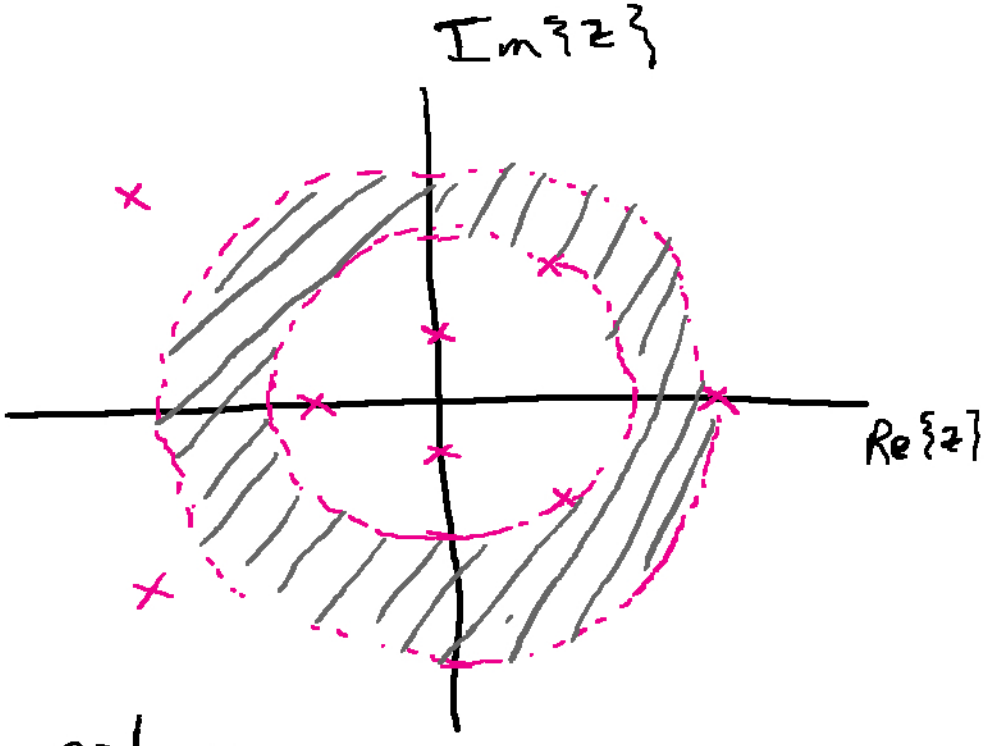


ROC extends in from smallest magnitude pole (except possibly $z = 0$)

7) Two-sided signal

4:55





ROC lies in a ring
bounded inside and outside by poles

Not all two-sided signals have z-transforms

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